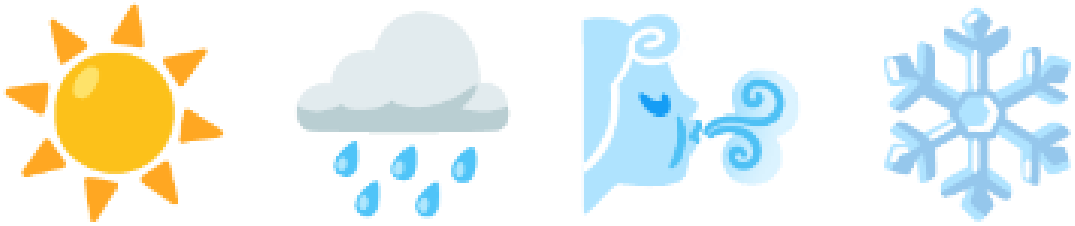
Little Lab Coats

🌤️ What Is Weather? 🌤️

Kindergarten Earth & Space Science — Unit 1: Weather & Seasons

📚 **Grade:** Kindergarten 🧭 **Unit:** K-ESS2 Unit 1 — Weather & Seasons 🔢 **Lesson:** 1 of 6 ⏱️ **Duration:** 30–45 minutes



What's the sky doing right now? That's weather!

📋 **NGSS Standard:**
K-ESS2-1 — Use and share observations of local weather conditions to describe patterns over time.

🎯 Learning Objective

Students will *identify* at least 4 types of weather and *record* daily weather observations for one week using a Weather Watcher Journal.

🔬 The Big Idea

Weather is what the atmosphere (the air around Earth) is doing right now — is it sunny? Rainy? Windy? Cold? Weather can change from hour to hour and day to day.

Climate is different — it's the average pattern of weather for a place over many, many years. Denver, Colorado has a semi-arid climate, which means it's usually dry — but any one day could still be rainy or snowy!

💡 **Key vocabulary:** weather, climate, atmosphere, observation, temperature, precipitation

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🌬️ Wind — Invisible but Powerful! 🌬️

Kindergarten Earth & Space Science — Unit 1: Weather & Seasons

📚 **Grade:** Kindergarten 🧭 **Unit:** K-ESS2 Unit 1 — Weather & Seasons 🔢 **Lesson:** 2 of 6 ⏱️ **Duration:** 30–45 minutes

You can't see wind — but you can feel it and measure it!

📋 **NGSS Standard:**
K-ESS2-1 — Use and share observations of local weather conditions to describe patterns over time.

🎯 Learning Objective

Students will *demonstrate* that wind is moving air and *measure* wind strength using a simple pinwheel or streamers.

🔬 The Big Idea

Wind is air in motion. You can't see air, but you can see what it does — it spins pinwheels, bends trees, and carries clouds across the sky. Wind is actually caused by the sun heating Earth's surface unevenly, creating areas of different air pressure. Air always moves from high pressure areas toward low pressure areas — and that movement is wind!

> 💡 **Key vocabulary:** wind, air, air pressure, force, pinwheel, anemometer

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🌧️ Rain, Snow, and the Water Cycle 💧

Kindergarten Earth & Space Science — Unit 1: Weather & Seasons

📚 **Grade:** Kindergarten 🧭 **Unit:** K-ESS2 Unit 1 — Weather & Seasons 🔢 **Lesson:** 3 of 6 ⏱️ **Duration:** 30–45 minutes

The same water has been cycling around Earth for billions of years!

📋 **NGSS Standard:**
K-ESS2-1 — Use and share observations of local weather conditions to describe patterns over time.

🎯 Learning Objective

Students will *describe* the water cycle and *demonstrate* evaporation and condensation using simple household materials.

🔬 The Big Idea: The Water Cycle

Water on Earth is always moving in a big loop called the **water cycle**. It evaporates from oceans and lakes into the air, forms clouds, falls back down as rain or snow, and flows back to the ocean — then the cycle starts all over again. The same water molecules have been cycling around Earth for billions of years!

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💡 **Key vocabulary:** water cycle, evaporation, condensation, precipitation, water vapor, clouds

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🌸 The Four Seasons 🍂

Kindergarten Earth & Space Science — Unit 1: Weather & Seasons

📚 **Grade:** Kindergarten 🧭 **Unit:** K-ESS2 Unit 1 — Weather & Seasons 🔢 **Lesson:** 4 of 6 ⏱️ **Duration:** 30–45 minutes

Spring, Summer, Fall, Winter — each season has its own special character!

📋 **NGSS Standard:**
K-ESS2-1 — Use and share observations of local weather conditions to describe patterns over time.

🎯 Learning Objective

Students will *sequence* the four seasons, *describe* characteristics of each, and *explain* how seasons affect plants and animals.

🔬 The Four Seasons

🌸 Spring

- Warmer after winter
- Flowers bloom
- Baby animals are born
- More rain
- Trees grow new leaves

☀️ Summer

- Hottest season
- Longest days (most daylight)
- Plants are fully leafy and green
- Fruits and vegetables grow
- Great time for long outdoor play

🍂 Fall (Autumn)

- Cooler weather
- Leaves change color
- Animals prepare for winter

❄️ Winter

- Coldest season
- Shortest days (least daylight)
- Many animals hibernate
- Trees lose their leaves

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- Animals prepare for winter
- Days get shorter

- Trees may lose all their leaves
- Snow or frost may appear

💡 **Key vocabulary:** season, spring, summer, fall/autumn, winter, temperature, pattern, tilt, Northern Hemisphere

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🐻 Weather & Animals — How Animals Respond 🦅

Kindergarten Earth & Space Science — Unit 1: Weather & Seasons

📚 **Grade:** Kindergarten 🧭 **Unit:** K-ESS2 Unit 1 — Weather & Seasons 🔢 **Lesson:** 5 of 6 ⏱️ **Duration:** 30–45 minutes

Animals can't put on a coat — but they have amazing ways to survive winter!

📋 **NGSS Standards:**
K-ESS2-1 — Use and share observations of local weather conditions to describe patterns over time.
K-ESS2-2 — Construct an argument supported by evidence for how plants and animals (including humans) can change the environment to meet their needs. *(Applies specifically to food caching and nest/lodge building.)*

🎯 Learning Objective

Students will *describe* at least 3 ways animals adapt their behaviors to seasonal weather changes.

🔬 The Big Idea

Animals cannot control the weather — so they've developed amazing **adaptations** (special behaviors and body features) that help them survive changing seasons. Some move to warmer places, some sleep through winter, some store food, and some change their appearance!

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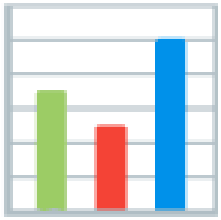
📊 Weather Patterns Over Time 🔍

Kindergarten Earth & Space Science — Unit 1: Weather & Seasons

📚 **Grade:** Kindergarten
🧭 **Unit:** K-ESS2 Unit 1 — Weather & Seasons
🔢 **Lesson:** 6 of 6 — Final Lesson!
⏱️ **Duration:** 30–45 minutes



Real scientists look at patterns to predict what comes next!

📋 **NGSS Standard:**
K-ESS2-1 — Use and share observations of local weather conditions to describe patterns over time.

🎯 Learning Objective

Students will *analyze* a week of weather data to *identify* patterns and *predict* tomorrow's weather.

🔬 The Big Idea

Scientists don't just observe — they look for **patterns**. When you collect weather data over days, weeks, and months, you start to see patterns: sunny days often come after rainy days, certain months are always colder, storms follow certain types of clouds. These patterns are how **meteorologists** (weather scientists) predict future weather!

💡 **Key vocabulary:** pattern, data, observation, prediction, meteorologist, bar graph

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